

Candidate Number _____

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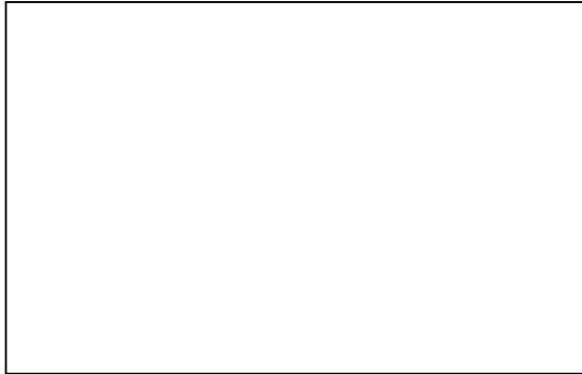
2017 International Biology Olympiad National Team Selection Camp Practical Test Questions

Experiment A: Zebrafish Dissection and Observation Answer Sheet

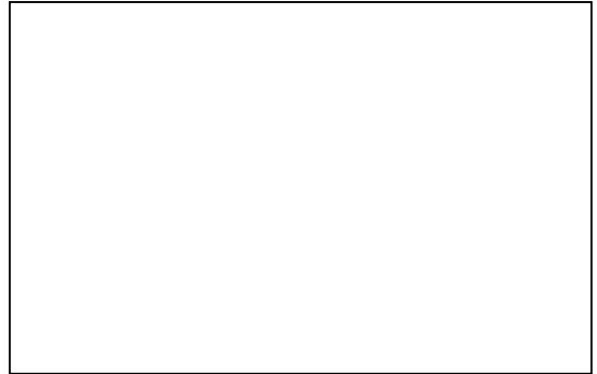
A. Observation of Zebrafish Scales (Total 15 points)

A1. ~~yyy yyy y yyy y~~ _____

Lateral line scales (4 minutes)



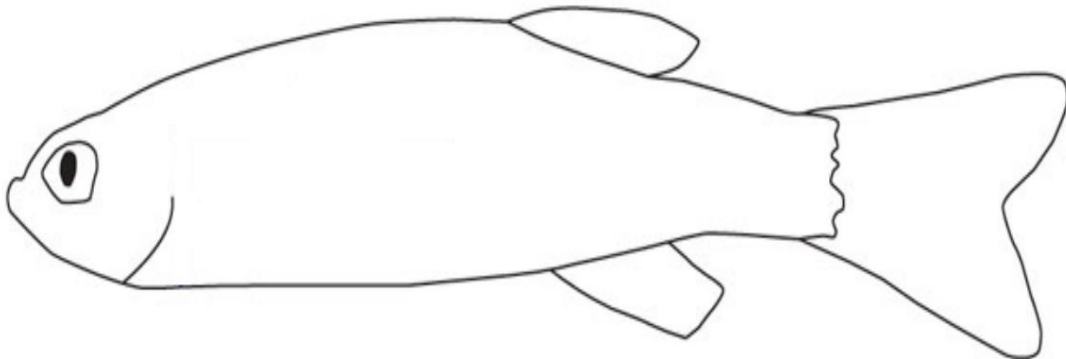
Non-lateral line scales (4 minutes)



Both lateral line scales and non-lateral line scales should be placed on a glass slide, sealed with a coverslip, and clearly marked with an oil-based pen for the invigilator to check. ~~(Scales should be placed on a glass slide, sealed with a coverslip, and clearly marked with an oil-based pen for the invigilator to check.)~~

A2. State the most significant difference between lateral line scales and non-lateral line scales. (4 points)

A3. Please draw the position of the lateral line on the zebrafish diagram below (2 points).



A4. The body length of a zebrafish is... _____

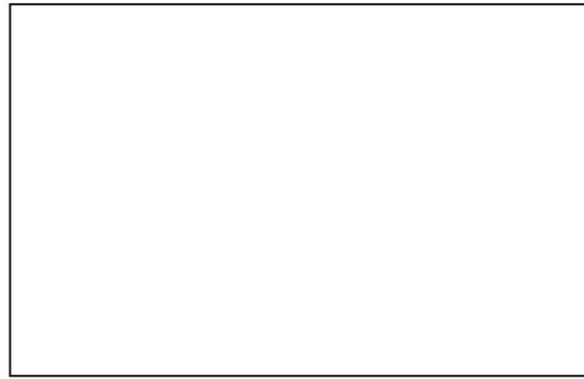
Total length in centimeters is _____

Centimeters (1 point)

B: Observation of the Internal Structure of Zebrafish **(35 points in total)** Candidate Number: _____

B1. Which is closer to the anal fin, the anus or the cloaca? (3 points) _____

B2. Draw the shape of the zebrafish otolith (the most obvious one) (10 points) (No points will be awarded for question B2 if a slide specimen is not provided)

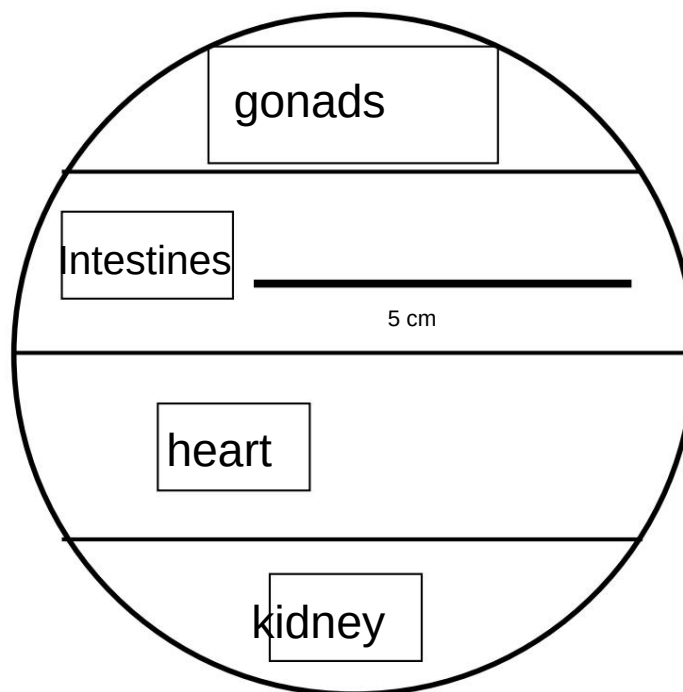


B3. Remove the following organs and place them in the designated area of a petri dish for scoring.

(Each section is worth 5 points, for a total of 20

points) All designated organ specimens must be placed in the designated location on the answer sheet for the invigilator to check. (No points will be awarded for questions that are not placed in the designated location on the answer sheet or for which no organ specimen is provided.)

Candidate Number: _____



B4. Based on the gonads observed after dissection, determine the sex of the zebrafish. (2 points) _____

Candidate Number (or Examination Number) _____

2017 International Biology Olympiad National Team Selection Camp Practical Test Questions

Animal development and physiology in test **site 2**

Practical Exercise A: Dissection and Observation of Zebrafish

I. Instructions for Experimental Apparatus:

A. Equipment 1 (Reusable)

1. Optical microscope	1 unit
2. Dissecting microscope	1 unit
3. Small dissection scissors	1 piece
4. Dissecting forceps	2
5. Dissecting needle	1 piece
6. Pin	4
7. Dissecting tray	1
8. Ruler (15 cm)	1 piece

Oil-based pen and leave on the lab table.

B. Equipment Two (One set updated for each person)

1. Zebrafish	1 tail
2. Petri dish	Group 1
3. Coverslip	1 box
4. Glass slide	4 pieces
5. Water (for washing)	1 bottle
6. Plastic graduated pipette (3 mL)	1 piece
7. Paper towels	1 pack
8. Gloves	1 pair

Disposable after use; each person uses a new set.

1. Please confirm that your candidate number is correct; if it is incorrect, please raise your hand and ask the teaching assistant for assistance.

2. Once the medicines and equipment on the table are used up, they will not be replenished.

3. This exam paper (including the cover and the question paper) consists of 3 pages and must be returned in its entirety when submitting the exam.

4. The total time allotted for answering questions **A** and **B** in this exam is **90** minutes. Please answer this question on the answer sheet.

Experiment A: Dissection and Observation of Zebrafish (50 points)

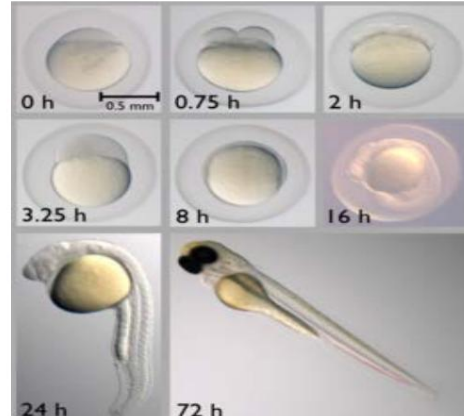
I. Introduction

Zebrafish (*Danio rerio*) is a tropical freshwater fish widely used in genetics and developmental biology.

The research focuses on this topic. Its advantages include ease of rearing and embryo transparency, facilitating observation of embryonic development and genetic testing.

Transgenic organisms, and whose internal organs, especially the digestive system, are very similar to humans, are well-suited for treating various intestinal diseases.

Research related to physics. It has now been officially designated as a model animal for pre-clinical trials.



Zebrafish appearance, female (top) and male (bottom); zebrafish fertilized eggs, embryos and fry.

II. Experimental Procedures and Problems

A. Observation of Zebrafish Scales (Total 15 points)

1. The skin of most fish is covered with scales that evolved from the dermis.

The fish has bony scales that protect its body. There are one to three scales on each side of its body, from the posterior edge of the gill cover to the tail.

The dotted-line structure formed by perforations or vessels on the scales is called the lateral line.

Scales through which the lateral line opening or canal passes are called lateral line scales. The number and morphology of lateral line scales are important for fish classification.

One of the bases for this is that its appearance is significantly different from that of ordinary scales (non-lateral line scales).

2. Use tweezers to remove one lateral line scale and one non-lateral line scale, observe them under a microscope, and draw their shapes.

Appearance and morphology (4 points each, 8 points in total) (A1), and compare the most significant differences between the two (4 points).

(A2)

Both lateral line scales and non-lateral line scales should be placed on a glass slide, sealed with a coverslip, and clearly labeled for the invigilator to check.

3. Please draw the position of the lateral line on the zebrafish diagram in the answer sheet (2 points) (A3).

4. Find the body length and total length of the zebrafish (1 point) (A4)

B. Observation of the internal structure of zebrafish (35 points in total)

1. Place the zebrafish belly-up in a petri dish and locate the anus or cloaca in front of the anal fin, noting their respective positions.
2. The cloaca of fish is a shared opening connecting the excretory and reproductive systems to the outside of the body. It is close to the anus. Based on the dissection results, determine which is the anus and which is the cloaca. (B.1) 3. Using dissecting scissors, cut open the zebrafish at the ventral side of the posterior edge of the gill cover to see its heart. Note the heart... Dirty areas.
4. Place the zebrafish with its left ventral side facing up on a dissecting dish. Using dissecting scissors, starting from the opening in step 2, cut along the midline of the abdomen towards the caudal fin until you reach the opening in front of the anal fin. Then, turn towards the dorsal spine and cut the body wall. Next, cut along the midline of the side of the body (approximately at the second blue stripe) to the posterior edge of the gill cover, and then cut along the posterior edge of the gill cover to the lower jaw. Carefully remove one side of the body wall and observe and mark the location and appearance of the following organs with different colored pins. 5. The inner ear is the auditory sensory organ of fish. There is one inner ear on each side of the head, and each inner ear consists of three otolith sacs, each containing an otolith. Otoliths are calcium carbonate stones that help form hearing in the inner ear. Carefully remove the most prominent otolith and observe it under a microscope, then draw its shape.
- (B2) 6. The zebrafish's gonads are located on the ventral side. Remove the gonads and place them in a petri dish on the answer sheet.
The corresponding location is for scoring. (B3)
7. The digestive tract of fish begins at the mouth, passes through the oral cavity, gill cavity, esophagus, intestine, and ends at the anus.
Cut the intestine completely and place it in the designated position on the corresponding answer sheet in a petri dish. Measure the total length of the intestine with a ruler. After measurement, please keep it in the corresponding position on the petri dish for scoring.
(B3)
8. The fish heart consists of several parts, including a ventricle, an atrium, a bulbus arteriosus, and sinus veins attached to the posterior end of the atrium. After completely removing the zebrafish heart, place it in the designated location on a petri dish. (B3)
9. From the cloaca upwards, you can find the ureter and kidney in sequence. The paired kidneys are closely attached to the dorsal side of the thoracic and abdominal cavities, appearing as long, dark red strips. Remove the fish's kidneys intact and place them in their corresponding positions on a petri dish. (B3) 10. Try to determine the sex based on the dissection results. (B4)

Each designated organ specimen must be placed in the designated location on the answer sheet for the invigilator to inspect.

(No points will be awarded for questions that are not placed in the designated area on the answer sheet or for which no organ specimen is provided).

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2017 International Biology Olympiad National Team Selection Camp Practical Test Questions

Second Examination **Room**

Animal development and physiology

Practice B

ÿ All the necessary items for the experiment are placed on the table. Please check them against the list below. If anything is missing, please raise your voice.

Tell the judges with your hand.

equipment	quantity
Mealworms	15

Precautions:

1. This exam paper (including cover and question paper) consists of 3 pages and must be returned in its entirety upon submission.
2. This exam includes Practice A and Practice B, with a total time limit of 90 minutes.
3. Please answer the questions on the answer sheet. You may write your answers on the back of the question paper, but please indicate and mark it.

Question Number

1. How to assess the preference of mealworm larvae for food groups A, B, and C, and its impact on development and egg production? Please explain the experimental design concept, including the control variables, the strain variables, prediction, experimental equipment, and the theoretically required data parameters. Please also draw a diagram of the experimental facilities and a flowchart of the experimental procedure.

